

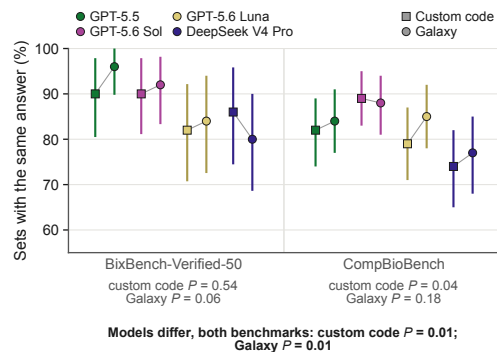
a What each model ran in Galaxy

Galaxy runs with at least one completed job in each family (%); UDTs were not offered on IWC

		BixBench				CompBio				IWC			
		10	38	51	33	19	28	44	33	0	30	30	33
Data handling	Tables and text	19	42	57	23	11	16	31	22	10	27	23	10
	Format conversion and upload	2	2	2	0	12	22	26	22	0	17	20	10
Scientific methods	Inspection and quality control	0	0	0	0	0	2	6	5	10	13	10	13
	Data retrieval	2	2	2	2	9	16	19	15	27	40	33	30
	Read processing and alignment	0	0	0	0	4	6	10	7	0	0	0	3
	Variant calling and annotation	3	6	9	5	2	3	4	5	20	17	20	10
	Expression and differential testing	0	0	0	0	2	4	5	7	3	3	0	10
	Single-cell and spatial analysis	2	2	2	2	9	15	27	23	3	10	7	13
	Genomic intervals and sequence features	12	16	21	17	0	0	0	0	0	0	0	0
	Phylogenetics	0	1	1	1	5	5	9	7	10	27	27	10
	Sequence search, taxonomy and assembly	0	0	0	0	1	1	2	2	3	10	10	3
	Peak calling and epigenomics	22	35	34	19	2	4	10	9	0	0	0	0
	Statistics, machine learning and enrichment	0	0	0	0	0	0	1	0	3	10	7	10
	Other methods	66	35	39	9	45	75	39	33	0	0	0	0
	UDT (agent-written code)												

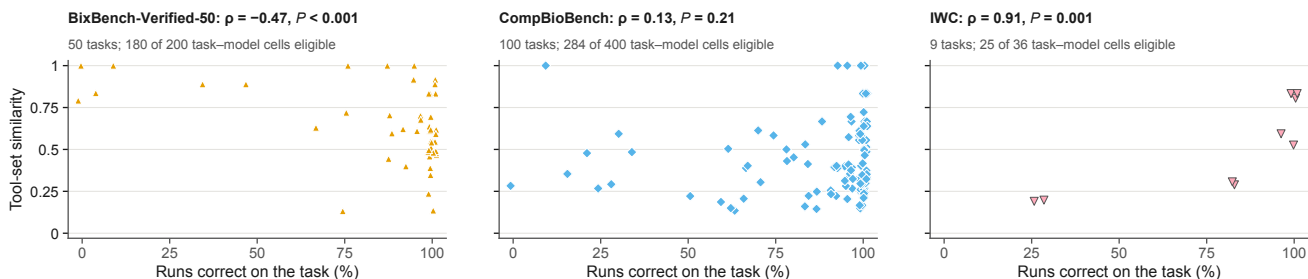
b Same model in all three runs

Replicate sets (one task, model and condition)



c Tool-set similarity and task accuracy

Galaxy; 1 = the same set of tools in all three runs (any UDT counted as one item; order, versions and parameters ignored). Models compared on the same task: Spearman $\rho = 0.07$, $P = 0.14$



d Replicate outcomes by held-out task difficulty

BixBench-Verified-50 and CompBioBench, both conditions; difficulty from the task's other 21 runs, so a set's own runs never define it

